

Course: B. Tech.

Branch : Mechanical Engineering

Semester : V

Subject Code & Name: BTMEC 501 & Heat Transfer

Max Marks: 60

Date: 07/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

- A) What is the thermal diffusivity? Enlist its two applications. (L3/CO1) 6
- B) Write is the difference between conduction and convection heat transfer? (L1/CO2) 6

List the assumptions of Fourier's Law.

- C) In a furnace, temp of hot gases is 2100 °C. Ambient temp is 40 °C. Heat flow by radiation from hot gases to inner surface of the wall is 23 kW/m². Convective heat transfer coefficient between hot gases and the inner surface of the wall is 12 W/m²K. Thermal conductance of the wall is 58 W/m²K. Heat flow by radiation from external surface of the wall to surroundings is 10 kW/m². Temp of inside surface of the wall is 900 °C. For the external surface of the wall, find surface temp and convective heat transfer coefficient. (L3/CO1) 6

Q.2 Solve Any Two of the following.

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- A) What is fin in heat transfer? Classify the fins based on operating condition and area. (L2/CO3) 6
- B) What are Fourier and Biot Numbers? What is physical significance of these numbers? (L1/CO2) 6
- C) Derive general heat conduction equation for Cartesian coordinate system. (L1/CO1) 6

Q. 3 Solve Any Two of the following.

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- A) Explain the concept of hydrodynamic and thermal boundary layers with reference to flow over a flat heated plate. (L1/CO1) 6
- B) Explain Grashoff Number, Nusselt Number and Prandtl Number. Also state the relationship between them. (L3/CO4) 6
- C) A circular disc insulated from other side of diameter of 25 cm is exposed to air at 20 °C. If the disc (Open Surface) is maintained at 120 °C, estimate heat transfer rate from it, when; a) Disc is kept horizontal with (open) hot surface facing upwards $N_u = 0.14 (Gr. Pr)^{0.334}$ (L3/CO4) 6

b) Disc is kept horizontal with (open) hot surface facing downwards

$$N_u = 0.27 (\text{Gr} \cdot \text{Pr})^{0.25}$$

Q.4 Solve Any Two of the following.

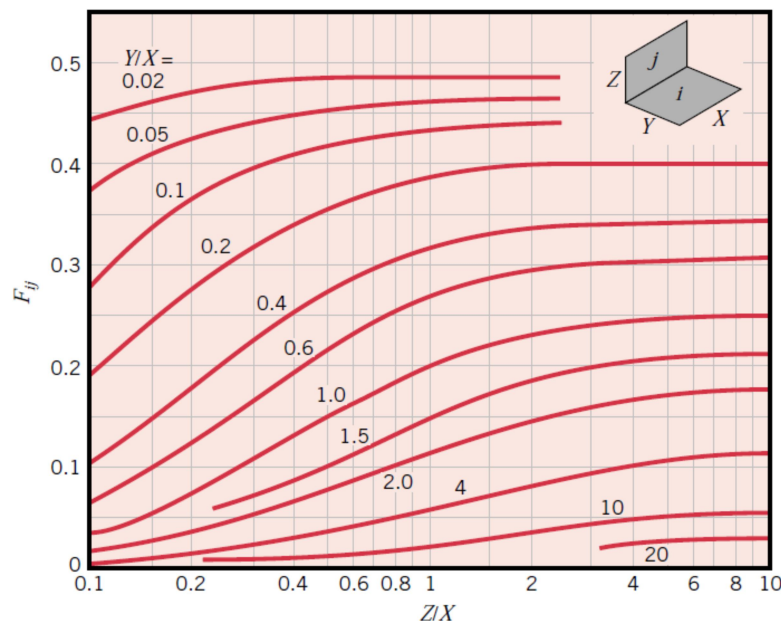
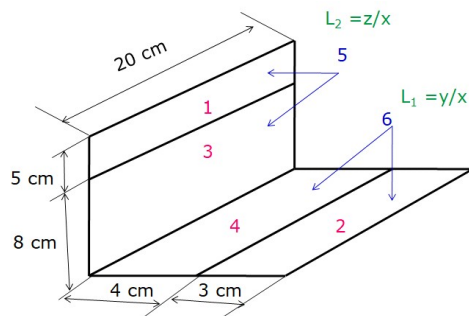
12

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|----|---|----------|---|
| A) | Derive the expression for LMTD method for an analysis of counter flow heat exchanger. | (L2/CO5) | 6 |
| B) | Write a short note on temperature distribution in parallel, counter and cross flow heat exchangers. | (L1/CO5) | 6 |
| C) | Water enters a counter flow double pipe heat exchanger at 38 °C flowing at the rate of 0.75 kg/s. It is heated by oil ($c_p = 1.884 \text{ kJ/kg} \cdot \text{K}$) flowing at the rate 1.5 kg/s from an inlet temperature of 116 °C. For an area of 13 m ² and overall heat transfer coefficient of 340 W/m ² .K, determine the total heat transfer rate? | (L3/CO5) | 6 |

Q. 5 Solve Any Two of the following.

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|----|--|----------|---|
| A) | Define shape factor. Explain relations/theorems of shape factor. | (L3/CO6) | 6 |
| B) | State and explain the Planck's law and Stefan Boltzmann law of radiation | (L1/CO6) | 6 |
| C) | Find F_{12} . | (L2/CO6) | 6 |



*** End ***